

Galois 1897 GPT Critical Edition

**Post-P13 adjudication of the 21 deferred tasks  reconstructed deep-replay
revision R2**

GPT critical-layer production

14 August 2026

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1 Galois 1897 GPT Critical Edition

Post-P13 adjudication of the 21 deferred tasks — reconstructed deep-replay revision R2

Build date: 2026-08-14. Revision R2 reconstructs the missing cumulative ZIP and closes the two formerly unresolved post-P13 task rows. This critical layer follows, and never rewrites, the frozen 78-page P13R diplomatic edition.

1.1 Layer contract

The 1897 French diplomatic transcription is immutable evidence. Every emendation, theorem repair, counterexample, qualification, witness collation, and provenance conclusion appears only in this appendix and is keyed to the stable IDs already present in the P13R handoff. The words **repaired** and **rejected** refer to the critical adjudication of a printed claim; they never describe a mutation of the diplomatic source.

Active repaired candidate SHA-256: 4909FB4852F2178C583CCD425451E8BD5C7B02BD2346807F79F617BB3F2E645F. P13R cold audit: PASS 74/74. The 78-page diplomatic PDF remains immutable. This R2 appendix corrects only the separate critical apparatus and adds independent-source evidence.

1.2 Master disposition of the 21 tasks

Task	Status	Result
POST-P13-A001	repaired	All 36 proved source-error records were adjudicated in a separate critical layer. Each has a proved repair, a qualified replacement, or a counterexample rejecting the printed claim; no diplomatic byte was changed.
POST-P13-A002	proved	A machine-readable graph records local and cross-work dependencies. All downstream outcomes are classified as survives, survives under added hypotheses, or blocked.
POST-P13-A003	proved	The dated prior-notice matrix covers all 36 errors and 15 variants. It distinguishes explicit notice, correct earlier witness, silent correction, candidate inaccessible errata source, and no notice found.
POST-P13-A004	proved	The $\mu > \nu$ argument survives: $F_{\{p^\nu\} \subseteq F_{\{p_\mu\}}}$ gives ν/μ , while the reciprocal rational dependence gives μ/ν ; hence $\mu = \nu$. The printed proof is compressed, not false.
POST-P13-A005	rejected	The degree-9/25-only converse classification is false. A solvable primitive affine group of degree 16 and order 288 with irreducible complement of order 18 is not one-dimensional semilinear.

Task	Status	Result
POST-P13-A006	repaired	The p=11 row is reconstructed with 4 in place of the duplicated 5. The six-pair matching has stabilizer order 60 and index 11 in $\text{PSL}(2,11)$.
POST-P13-A007	proved	The 1846 witness prints réduction, the 1897 witness prints équation, and Tannery explicitly collates the substitution in 1908. The change affects semantic scope, not a formula.
POST-P13-A008	repaired	A 600 ppi replay corrects the apparatus transcription from $F'F''$ to $E'F''$. The manuscript determinant $E'F'' - E''F' = (\pi/2)\sqrt{-1}$ is proved equivalent to the printed Legendre relation under the classical real/imaginary period normalization; Tannery 1908 is the earliest explicit notice located.
POST-P13-A009	proved	The 1846 witness has cette fonction at W09-PE001. W09-PE002 is diagnosed from the 1897 form propriété itself; no 1846 comparison reading for that exact token was established. No explicit itemized notice was located for either record.
POST-P13-A010	repaired	Lemma III is repaired by a gcd/Bézout argument in $K(V)[X]$. This closes Proposition I and later rational-resolvent uses.
POST-P13-A011	repaired	Proposition II is replaced by the field-intersection/index theorem, and Proposition III by the normal-intersection proof. Composite-degree overstatement is rejected by an exact degree-4 example.
POST-P13-A012	repaired	The fixed Fourier coefficient may vanish; the repair chooses a nonzero nontrivial coefficient guaranteed by Fourier inversion. The radical-tower step then survives.
POST-P13-A013	repaired	The Kummer/normality hypotheses and the word nonidentity are restored. Proposition VIII survives; the unqualified Proposition VI statement is refuted by x^3-2 .
POST-P13-A014	proved	The W09 corpus was collated against 1846, Tannery 1908, the 1962/1976 critical-edition records, Neumann 2011/2013, and Dicker 2026. Prior art and no-notice findings are separated.
W10-DA001	repaired	The printed unique-block proof is incomplete. A complete solvable-primitive proof uses a minimal normal elementary-abelian subgroup, whose orbits are blocks; it is transitive, regular, and yields affine degree p^d .

Task	Status	Result
W10-DA002	repaired	Affine linearity is valid under the inherited finite solvable primitive hypotheses. The unqualified primitive reading is false, for example for $S_{\{p^2\}}$ in its natural primitive action.
W10-DA003	rejected	The premise is rejected. Nineteenth-century conjugate/conjoined substitution terminology does not imply pairwise commutation; explicit conjugate projective maps on $P^1(F_7)$ fail to commute.
W10-DA004	repaired	All six printed notation/formula repairs and both identity qualifications were propagated. The affine order counts survive; the terminal PDF089 p=3 inference remains blocked by W10-DA003.
W11-DA001	proved	The earliest securely dated correction found is the Harvard-copy Wikisource page printing v, last modified 2018-07-22. Earlier item-level errata in the 1976 list could not be inspected.
W11-DA002	proved	The official ETH Rar 4575 PDF and OCR were retrieved. PDF page 79 gives the five-digit printer job number 24572; character-by-character image replay and the independent OCR agree. The earlier four-digit premise is rejected.
W11-DA003	proved	Replay proves the terminal topology: PDF091 predecessor blank, PDF092 historical contents, PDF093 colophon, and three distinct trailing blanks PDF094-096.

1.3 Critical errata catalogue: all 36 frozen IDs

1.3.1 W02-PE001 — PDF 025 / L0024

Adjudication: typographical_repair_proved

Diplomatic printed form: déveveloppent

Critical repair or replacement: développent

Exact hypotheses: none

Proof certificate: French morphology and the immediate Picard-context establish that the duplicated syllable in déveveloppent is typographical. Picard's 1897 Introduction has no 1846 parallel witness.

Propagation: The Introduction's meaning is unchanged.

Historical result: No earlier witness exists for Picard's 1897 Introduction. Modern comparison transcriptions normalize the form, but no explicit itemized erratum was found in the searched corpus.

Sources: SRC-LOCAL-1897; SRC-WIKI-1897. The diplomatic layer is unchanged.

1.3.2 W03-PE001 — PDF 035 / L0034

Adjudication: formula_repair_proved

Diplomatic printed form: $a + 1/(-1/B) = a - B$; $a - B$

Critical repair or replacement: Replace the printed letter a by p in the conjugate expression, giving p-B.

Exact hypotheses: the article's established integer-part symbol p

Proof certificate: From $x=p+1/A$ and the conjugate $A \mapsto -1/B$, direct substitution gives $p+1/(-1/B)=p-B$.

Propagation: The corrected conjugate and the following interval argument become coherent; no later work cites the misprinted letter.

Historical result: No prior explicit notice found in the searched corpus.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.3 W03-PE002 — PDF 035 / L0034

Adjudication: sign_repair_proved

Diplomatic printed form: compris entre 0 et 1

Critical repair or replacement: Replace “between 0 and 1” by the inequality $-1 < p-B < 0$.

Exact hypotheses: B has integer part p and fractional part r in (0,1)

Proof certificate: Write $B=p+r$ with $0 < r < 1$. Then $p-B=-r$, hence $-1 < -r < 0$.

Propagation: Repairs the sign used in the conjugate-root discussion; no cross-work dependency.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.4 W03-PE003 — PDF 037 / L0036

Adjudication: formula_repair_proved

Diplomatic printed form: $x = 3 - 1/A$ (printed as the full staircase reciprocal of the positive y expansion)

Critical repair or replacement: Replace the second value $3-1/A$ by $3-A$.

Exact hypotheses: A is the positive root of $3A^2-2A-3=0$

Proof certificate: For $A=(1+\sqrt{10})/3$, exact reduction gives $3(3-A)^2-16(3-A)+18=0$, whereas the printed value has nonzero residual.

Propagation: The final periodic continued fraction evaluates to $3-A$; the theorem survives.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.5 W04-SE001 — PDF 038 / L0037

Adjudication: missing_hypothesis_repaired

Diplomatic printed form: Soient F_x et f_x deux fonctions quelconques données; on aura, quels que soient x et h , [quotient]

Critical repair or replacement: Restrict the quotient to pairs (x,h) for which $f(x+h) \neq f(x)$.

Exact hypotheses: $f(x+h) \neq f(x)$

Proof certificate: For constant f the denominator vanishes for every x,h , so the printed universal quotient is undefined.

Propagation: Only the domain of the displayed quotient changes.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897; SRC-NUMDAM-1846. The diplomatic layer is unchanged.

1.3.6 W04-SE002 — PDF 038 / L0037

Adjudication: printed_claim_rejected_counterexample

Diplomatic printed form: ce qui démontre, a priori, l'existence des fonctions dérivées

Critical repair or replacement: Delete the asserted a-priori existence of derivative functions; replace it by an explicit differentiability or quotient-limit hypothesis.

Exact hypotheses: existence of the relevant two-sided limit, or differentiability assumptions

Proof certificate: $F(t)=|t|$, $f(t)=t$ at $x=0$ yields $|h|/h$ with one-sided limits 1 and -1.

Propagation: The following mean-value discussion can proceed only after the new hypothesis.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.7 W04-SE003 — PDF 038 / L0037

Adjudication: missing_hypothesis_repaired

Diplomatic printed form: donc on doit avoir aussi $P = \phi(k)$

Critical repair or replacement: Require ψ to be injective on the relevant range, or select and state a single inverse branch ϕ .

Exact hypotheses: injectivity or a specified inverse branch

Proof certificate: $\psi(0)=\psi(1)=0$ makes $\phi(\psi(P))=P$ impossible simultaneously for $P=0,1$.

Propagation: The inversion step survives under the stated restriction.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.8 W04-SE004 — PDF 038 / L0037

Adjudication: missing_hypothesis_repaired

Diplomatic printed form: à moins qu'elle ne reste constante entre ces limites [...] cette fonction aura, entre x et $x+h$, un ou plusieurs maxima et minima

Critical repair or replacement: Assume continuity on the closed interval; conclude at least one interior extremum unless the function is constant. Claim both a maximum and a minimum only when values occur on both sides of the common endpoint value.

Exact hypotheses: continuity plus the indicated sign condition for two extrema

Proof certificate: Without continuity, equal endpoint values do not imply attainment: $F(0)=F(1)=0$ and $F(t)=t$ for $0<t<1$ has no maximum.

Propagation: Supplies the extremum needed for the mean-value argument; the stronger two-extrema wording is qualified.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.9 W05-SE001 — PDF 040-041 / L0039-L0040

Adjudication: printed_classification_rejected_counterexample

Diplomatic printed form: A part les cas mentionnés ci-dessous [...] deux quelconques de ses racines étant connues, les autres s'en déduisent rationnellement.

Critical repair or replacement: Replace the finite exception list by the modern affine theorem: a finite solvable primitive group has a regular elementary-abelian socle and an irreducible point stabilizer; one-dimensional semilinearity is an additional restriction.

Exact hypotheses: finite solvable primitive permutation group

Proof certificate: A degree-8 $\text{AGL}(1,8)$ example already refutes the printed list. Independently, $V=F_2^4$ with $H=(C_3 \times C_3) \rtimes C_2$ of order 18 gives a solvable primitive degree-16 group not contained in $\Gamma L(1,16)$, since 18 does not divide 60.

Propagation: The later degree-9/25-only claim at PDF051 is also rejected; see POST-P13-A005.

Historical result: No prior explicit notice of this exact counterexample was found; modern affine-group theory supplies the replacement framework.

Sources: SRC-LOCAL-1897; SRC-AFFINE-PRIMITIVE; SRC-LI-2003. The diplomatic layer is unchanged.

1.3.10 W05-SE002 — PDF 041 / L0040

Adjudication: printed_claim_rejected_source_internal_correction

Diplomatic printed form: Au contraire, pour des degrés supérieurs, les équations modulaires ne peuvent s'abaisser.

Critical repair or replacement: Qualify the impossibility statement by recording the $p=7$ and $p=11$ exceptional lowerings.

Exact hypotheses: none beyond the modular-equation setting

Proof certificate: Liouville's footnote in the 1846 and 1897 printings explicitly records these exceptions; exact subgroup indices 7 and 11 corroborate the mechanism.

Propagation: Feeds directly into W08-PE001's $p=11$ matching.

Historical result: Earliest explicit notice is Liouville's 1846 footnote, reprinted in 1897.

Sources: SRC-LOCAL-1846; SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.11 W06-PE001 — PDF 042 / L0041

Adjudication: coherent_formula_reconstruction_proved_nonunique

Diplomatic printed form: $x = \psi x = \text{nth_root}(X / (x^n/Y))$

Critical repair or replacement: In the critical layer use the symmetric candidate $\psi(x) = (Y/(X/x^n))^{1/n}$; do not assert that this is the unique historical correction.

Exact hypotheses: X, Y and x nonzero where the quotient is formed, with compatible fixed radical branches; further monotonicity is required for bracketing

Proof certificate: Exact polynomial examples prove that the printed map is not equivalent to $X=Y$ and can lie on the same side of x as the first map. The symmetric candidate is algebraically equivalent to $X=Y$ wherever denominators and the selected radical branches are defined.

Propagation: The algebraic fixed-point equivalence is restored by this critical reconstruction, but the bracketing assertion still requires branch and monotonicity hypotheses; no later work uses the formula.

Historical result: No prior explicit notice or uniquely documented historical emendation was found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.12 W06-SE001 — PDF 043 / L0042

Adjudication: missing_hypothesis_repaired

Diplomatic printed form: $nX - xX' > 0, nY - xY' > 0$; take $k = (nX - xX')$ at $x=1$

Critical repair or replacement: For $X = \sum c_j x^j$ with $c_j \geq 0$, require at least one $c_j > 0$ for $j < n$; analogously for Y .

Exact hypotheses: $x > 0$, nonnegative coefficients, and a positive coefficient below degree n

Proof certificate: $nX - xX' = \sum_{j < n} (n-j)c_j x^j$. On $x > 0$ it is strictly positive exactly when a lower-degree coefficient is positive.

Propagation: The parameter k is then nonzero and the rational iteration is defined.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.13 W07-SE001 — PDF 046 / L0045

Adjudication: printed_example_rejected

Diplomatic printed form: $x^2 + x + 1 = 0 \pmod{2}$ is offered as an example showing the roots are not expressible by radicals because the ordinary quadratic formula reduces to $0/0$.

Critical repair or replacement: Remove the example as evidence against radical expressibility; retain only the observation that the characteristic-zero quadratic formula degenerates.

Exact hypotheses: none

Proof certificate: A root α of $x^2 + x + 1$ over F_2 satisfies $\alpha^3 = 1$ and is itself a nontrivial cube root of unity.

Propagation: No later finite-field construction depends on the invalid example.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.14 W07-SE002 — PDF 049 / L0048

Adjudication: missing_hypothesis_repaired

Diplomatic printed form: $a^{m(p-1)} = 1$ and $a_1^{m(p-1)} = 1$ are used without a nonzero qualification.

Critical repair or replacement: Add $a \neq 0$ and $a_1 \neq 0$ modulo p before applying Fermat's theorem.

Exact hypotheses: nonzero coefficients

Proof certificate: Fermat's relation $u^{p-1} = 1$ holds on $F_p^\times$. Exact enumeration confirms failure precisely when one selected coefficient is zero.

Propagation: The worked choice $a = -1, a_1 = 1$ satisfies the repair, so its primitive-element conclusion survives.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.15 W07-SE003 — PDF 050 / L0049

Adjudication: algorithm_repaired

Diplomatic printed form: The same derivative-gcd method as for ordinary equations is said always to remove repeated roots.

Critical repair or replacement: Add the characteristic- p branch: if $F'=0$, take a p -th root of F and recurse; otherwise perform squarefree factorization with $\gcd(F,F')$.

Exact hypotheses: perfect coefficient field or explicit p -th-root availability for the extraction step

Proof certificate: For $F=x^3-1$ over F_3 , $F'=0$ and $F/\gcd(F,F')=1$ loses the root.

Propagation: Repairs the repeated-root algorithm; later finite-field factor selection survives.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.16 W07-SE004 — PDF 050 / L0049

Adjudication: formula_repair_proved

Diplomatic printed form: Integral solutions are obtained from $\gcd(F, x^{p-1}-1)$.

Critical repair or replacement: Use $\gcd(F, x^{p-x})$ to include every residue, or test $x=0$ separately before using $x^{p-1}-1$.

Exact hypotheses: none

Proof certificate: $F=x$ has the integral solution 0, but $\gcd(x, x^{p-1}-1)=1$.

Propagation: The residue-root extraction algorithm then covers zero and nonzero roots.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.17 W07-SE005 — PDF 050 / L0049

Adjudication: formula_repair_proved

Diplomatic printed form: Solutions of order ν are said to be given by $\gcd(F, x^{p\nu-1}-1)$.

Critical repair or replacement: For exact degree ν , remove all proper-subfield factors; equivalently use the product of monic irreducibles whose degrees equal ν , obtainable by Möbius inversion from $x^{p^d}-x$.

Exact hypotheses: finite fields and the standard subfield lattice

Proof certificate: $x^{p\nu-1}-1$ contains every nonzero element of F_{p^d} for $d|\nu$; for $p=3, \nu=2$, the degree-one element 1 is already a root.

Propagation: The general factor-selection method survives after exact-degree isolation.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.18 W08-PE001 — PDF 058 / L0057

Adjudication: witness_supported_formula_repair

Diplomatic printed form: infinity, 1, 3, 5, 5, 9

Critical repair or replacement: Replace the second 5 by 4, giving upper row $\infty, 1, 3, 4, 5, 9$.

Exact hypotheses: the displayed projective action

Proof certificate: The corrected pairs partition $P^1(F_{11})$; exact enumeration gives $|\text{PSL}(2, 11)|=660$ and matching stabilizer 60, hence index 11.

Propagation: Restores the $p=11$ modular lowering; no later proof in the volume depends on the duplicate.

Historical result: The 1846 publication already prints 4, but no explicit erratum was found; earliest attested correct witness is 1846.

Sources: SRC-LOCAL-1846; SRC-LOCAL-1897; SRC-LOCAL-1908. The diplomatic layer is unchanged.

1.3.19 W08-PE002 — PDF 058 / L0057

Adjudication: witness_supported_lexical_repair

Diplomatic printed form: En toute rigueur, cette équation n'est pas possible dans les cas plus élevés.

Critical repair or replacement: Replace équation by réduction in the critical layer.

Exact hypotheses: none

Proof certificate: The 1846 publication has réduction; Tannery's 1908 collation identifies the 1897 substitution. Semantically, réduction names the degree-lowering operation under discussion, whereas équation does not.

Propagation: No mathematical formula changes; the scope of the impossibility sentence is restored.

Historical result: Correct in 1846; explicitly noticed by Tannery in 1908.

Sources: SRC-LOCAL-1846; SRC-LOCAL-1897; SRC-LOCAL-1908. The diplomatic layer is unchanged.

1.3.20 W09-PE001 — PDF 065 / L0064

Adjudication: witness_supported_lexical_repair

Diplomatic printed form: en permutant dans celle fonction

Critical repair or replacement: Replace celle fonction by cette fonction.

Exact hypotheses: none

Proof certificate: The demonstrative must agree with fonction and the 1846 witness prints cette.

Propagation: No mathematical dependency.

Historical result: Correct in 1846; no explicit later notice found.

Sources: SRC-LOCAL-1846; SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.21 W09-PE002 — PDF 071 / L0070

Adjudication: typographical_repair_proved

Diplomatic printed form: propriété

Critical repair or replacement: Restore the accent: propriété.

Exact hypotheses: none

Proof certificate: The 1897 form propriété is a missing-diacritic typographical error under standard French orthography. The local collation did not establish an 1846 comparison reading for this exact token.

Propagation: No mathematical dependency.

Historical result: No explicit prior notice or securely established earlier corrected witness was found in the searched corpus.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.22 W09-SE001 — PDF 065-066 / L0064-L0065

Adjudication: proof_gap_repaired

Diplomatic printed form: The unique common root is said to be “sought”, and the root is therefore declared rational in V .

Critical repair or replacement: In $K(V)[X]$, prove $\gcd(P,Q)=X-a_i$ and use a Bézout identity to express a_i rationally in V .

Exact hypotheses: separability/distinct roots and uniqueness of the selected common root

Proof certificate: Uniqueness of the common root makes the monic gcd linear; Euclid gives $U P + V Q = X - a_i$, hence a_i lies in $K(V)$.

Propagation: Closes Lemma III, Proposition I, and every later construction that treats roots as rational functions of the resolvent V .

Historical result: Galois's own insufficiency note for Lemma III is reported explicitly by Tannery in 1908. No earlier published Bézout repair was identified in the searched corpus.

Sources: SRC-LOCAL-1908. The diplomatic layer is unchanged.

1.3.23 W09-SE002 — PDF 069-070 / L0068-L0069

Adjudication: theorem_repaired

Diplomatic printed form: After adjoining a root of an irreducible auxiliary equation, the group is unchanged or divides into p equal groups, although the prime-degree phrase defining p was deleted.

Critical repair or replacement: Let L/K be the original splitting field, $M=K(r)$, $E=L \cap M$, $H=\text{Gal}(L/E)$. Then $[G:H]=[E:K]$ divides $[M:K]$. Recover the 1-or- p dichotomy only when $[M:K]=p$ is prime.

Exact hypotheses: finite separable splitting field; prime degree only for the dichotomy

Proof certificate: The Galois correspondence gives H and the compositum degree formula gives $[E:K][M:K]$. The

degree-4 element $\sqrt{2}+\sqrt{3}$ yields an index-2 counterexample to a composite-degree dichotomy.

Propagation: Supplies the correct subgroup decomposition used in Proposition III and the radical tower.

Historical result: Tannery 1908 records the deleted prime-degree wording and incompleteness; Dicker 2026 is a modern dedicated reconstruction.

Sources: SRC-LOCAL-1908; SRC-DICKER-2026. The diplomatic layer is unchanged.

1.3.24 W09-SE003 — PDF 070-071 / L0069-L0070

Adjudication: proof_gap_repaired

Diplomatic printed form: The revised theorem is printed without its proof (“On trouvera la démonstration”).

Critical repair or replacement: If M/K is a splitting field, $E=L\cap M$ is Galois over K , so $H=\text{Gal}(L/E)$ is normal in G . Therefore all conjugate subgroups coincide after adjoining all auxiliary roots.

Exact hypotheses: both L/K and M/K finite Galois

Proof certificate: Intersections of finite normal separable extensions are normal and separable.

Propagation: Repairs Proposition III and every later use of a common normal subgroup.

Historical result: Tannery 1908 records the revised statement, erased predecessor proof, and Liouville interpolation. No prior published normal-intersection proof for this exact passage was identified in the searched corpus.

Sources: SRC-LOCAL-1908. The diplomatic layer is unchanged.

1.3.25 W09-SE004 — PDF 073 / L0072

Adjudication: proof_gap_repaired

Diplomatic printed form: The proof adjoins the p -th root of one specifically displayed Fourier combination raised to p .

Critical repair or replacement: Define all Fourier coefficients $R_m=\sum \zeta^{\{m_j\}}\theta_j$ and choose a nonzero coefficient with $m\neq 0$; do not privilege R_1 .

Exact hypotheses: p invertible in the base field and a primitive p -th root of unity available

Proof certificate: A cyclic example has $R_1=0$ and $R_2=3t$. Fourier inversion shows all nontrivial coefficients cannot vanish unless the orbit is constant.

Propagation: The selected nonzero R_m has invariant p -th power and supports the same radical adjunction.

Historical result: No prior explicit notice found in the searched corpus.

Sources: SRC-LOCAL-1897; SRC-NEUMANN-2013. The diplomatic layer is unchanged.

1.3.26 W09-SE005 — PDF 075 / L0074

Adjudication: missing_hypothesis_repaired

Diplomatic printed form: An irreducible prime-degree equation is said not to become reducible unless its group reduces to one permutation.

Critical repair or replacement: Require the radical extension to be cyclic Galois of prime degree in the inherited Kummer setting. Then reducibility of a prime-degree irreducible polynomial forces the relevant normal subgroup to be trivial, hence complete splitting.

Exact hypotheses: base contains the needed roots of unity; radical extension irreducible, cyclic, Galois, and prime degree

Proof certificate: x^3-2 over $\mathbb{Q}$ becomes reducible over $\mathbb{Q}(\sqrt[3]{2})$ but does not split. Under normal prime-degree Kummer hypotheses, subgroup orbits form a block system, so nontransitive normal subgroup is trivial.

Propagation: Propositions VI-VII survive only within this Kummer context.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897; SRC-NEUMANN-2013. The diplomatic layer is unchanged.

1.3.27 W09-SE006 — PDF 078 / L0077

Adjudication: missing_qualifier_repaired

Diplomatic printed form: The affine substitution $x_k\mapsto x_{\{ak+b\}}$ is said never to leave two letters in place, with no exclusion of the identity.

Critical repair or replacement: Insert nonidentity before the affine fixed-point claim.

Exact hypotheses: none

Proof certificate: For $k\mapsto ak+b$, the identity fixes all p points; a nonidentity map fixes zero points if $a=1, b\neq 0$ and exactly

one if $a \neq 1$.

Propagation: Proposition VIII's two-root criterion survives because only nonidentity elements are relevant.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.28 W10-PE001 — PDF 082 / L0081

Adjudication: witness_supported_index_repair

Diplomatic printed form: $a_{\{k_1, k_1, k_3, \dots, k_\mu\}}$

Critical repair or replacement: Replace the repeated k_1 by k_2 .

Exact hypotheses: none

Proof certificate: The declared independent indices require $k_1, k_2, \dots$ and Tannery explicitly records the correction.

Propagation: Restores the coordinate tuple; subsequent affine construction survives.

Historical result: Explicitly corrected by Tannery in 1908.

Sources: SRC-LOCAL-1897; SRC-LOCAL-1908. The diplomatic layer is unchanged.

1.3.29 W10-PE002 — PDF 082 / L0081

Adjudication: witness_supported_index_repair

Diplomatic printed form: $\psi(k)_2$

Critical repair or replacement: Read $\psi(k_2)$, not $\psi(k)_2$.

Exact hypotheses: none

Proof certificate: Coordinate parallelism and Tannery's correction agree.

Propagation: Restores the second coordinate function.

Historical result: Explicitly corrected by Tannery in 1908.

Sources: SRC-LOCAL-1897; SRC-LOCAL-1908. The diplomatic layer is unchanged.

1.3.30 W10-PE003 — PDF 083 / L0082

Adjudication: witness_supported_index_repair

Diplomatic printed form: $a_{\{mk_1+n, k_2\}}$

Critical repair or replacement: Read $a_{\{mk_1+n, k_2\}}$.

Exact hypotheses: none

Proof certificate: The transformation changes the first coordinate and leaves the second fixed; Tannery gives the separated pair.

Propagation: The one-coordinate affine action and later two-coordinate action remain coherent.

Historical result: Explicitly corrected by Tannery in 1908.

Sources: SRC-LOCAL-1897; SRC-LOCAL-1908. The diplomatic layer is unchanged.

1.3.31 W10-PE004 — PDF 083 / L0082

Adjudication: context_forced_index_repair

Diplomatic printed form: $a_{\{k_2, k_3\}}$

Critical repair or replacement: Read $a_{\{k_1, k_2\}}$.

Exact hypotheses: none

Proof certificate: Only k_1, k_2 have been introduced in the section, and the target has those two coordinates.

Propagation: Restores a map on the stated domain.

Historical result: No explicit prior notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.32 W10-PE005 — PDF 084 / L0083

Adjudication: witness_supported_formula_repair

Diplomatic printed form: $a_{\{m_1k_1+n_1k+\alpha_1m_2k_1+n_2k_2+\alpha_2\}}$

Critical repair or replacement: Reconstruct formula (A) as $a_{\{m_1k_1+n_1k_2+\alpha_1, m_2k_1+n_2k_2+\alpha_2\}}$.

Exact hypotheses: invertible linear part where a permutation is intended

Proof certificate: It is the affine form forced by the finite-difference condition and is explicitly supplied by Tannery.

Propagation: The affine-linearity argument can then be stated correctly under W10-DA002's inherited solvability hypotheses.

Historical result: Explicitly corrected by Tannery in 1908.

Sources: SRC-LOCAL-1897; SRC-LOCAL-1908. The diplomatic layer is unchanged.

1.3.33 W10-PE006 — PDF 088 / L0087

Adjudication: formula_repair_proved

Diplomatic printed form: $(rk+s)k-m(mk+n)=0$

Critical repair or replacement: Remove the extra factor m: $(rk+s)k-(mk+n)=0$.

Exact hypotheses: projective denominator treated in P^1

Proof certificate: This is the direct fixed-point equation of $T(k)=(mk+n)/(rk+s)$; its discriminant is the next printed expression $(m-s)^2+4nr$.

Propagation: Restores the fixed-point/support trichotomy; the $p=3$ terminal argument still fails independently under W10-DA003.

Historical result: The same error occurs in 1846; no explicit erratum was found.

Sources: SRC-LOCAL-1846; SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.34 W10-SE001 — PDF 086 / L0085

Adjudication: missing_qualifier_repaired

Diplomatic printed form: m est nécessairement <2 et ... 0 ou 1

Critical repair or replacement: Insert nonidentity before asserting fixed-space dimension <2 .

Exact hypotheses: none

Proof certificate: A nonidentity affine map has empty fixed set or an affine fixed set of dimension 0 or 1; the identity has dimension 2.

Propagation: The divisibility bound survives; the identity is handled separately.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.35 W10-SE002 — PDF 087 / L0086

Adjudication: count_repaired

Diplomatic printed form: dans ces substitutions, aucune lettre ne reste à la même place, et elles sont au nombre de p^2-1

Critical repair or replacement: Replace p^2-1 by p^2-2 in the count of nonidentity scalars moving every nonzero letter; retain subgroup order p^2-1 .

Exact hypotheses: none

Proof certificate: The scalar group contains identity; every other scalar fixes only zero.

Propagation: The final affine-group order $p^2(p^2-1)(p^2-p)$ is unchanged.

Historical result: No prior explicit notice found.

Sources: SRC-LOCAL-1897. The diplomatic layer is unchanged.

1.3.36 W11-PE001 — PDF 092 / L0091

Adjudication: page_reference_repair_proved

Diplomatic printed form: 1

Critical repair or replacement: Replace the table-of-contents page reference 1 by roman v in the critical layer.

Exact hypotheses: none

Proof certificate: The Introduction occupies logical folios v-x by the primary page map; the 1897 table prints Arabic 1.
Propagation: Only the critical navigation reference changes.
Historical result: A Harvard-copy Wikisource transcription silently prints v and was last modified 2018-07-22; this is the earliest securely dated notice found.
Sources: SRC-LOCAL-1897; SRC-WIKI-T0C; SRC-ETH-1897. The diplomatic layer is unchanged.

1.4 The 21 task certificates

1.5 POST-P13-A001 — complete errata repair

Status: repaired

This certificate belongs to the critical layer. The P13R diplomatic transcription and its 78-page PDF are immutable. No wording below authorizes a silent change to that layer. Coordinates are physical PDF/JP2 coordinates from the frozen map.

Theorem. Every one of the 36 records in `CRITICAL_ERRATA_CATALOGUE.csv` has been closed by exactly one of the following critical actions: a typographical or formula repair proved from the 1897 context and another witness; an added hypothesis proved necessary and sufficient for the local step; a replacement proof; or a counterexample rejecting the printed universal claim.

Proof. The catalogue is an exact join with the frozen 36-row source-error ledger. Its `error_id` set is identical, every row states `diplomatic_layer_mutated=no`, and each row has a nonempty adjudication, repair, proof, propagation outcome, and prior-notice outcome. Exact symbolic and finite-group checks are in `qa/MATHEMATICAL_CHECKS.json`. No record has been deleted, merged, or renumbered.

The principal nonlocal replacements are: the affine primitive theorem for W05-SE001 and POST-P13-A005; the characteristic-p squarefree and exact-degree algorithms for W07-SE003–005; the field-intersection, Bézout, Fourier, and Kummer repairs for W09-SE001–006; and the minimal-normal-subgroup affine proof for W10-DA001–002. The remaining entries are local textual, sign, formula, index, or identity corrections.

Conclusion. The errata layer is complete as an adjudication of the frozen 36 records. It is not a translation and it does not assert that every diagnosis is historically new.

1.6 POST-P13-A002 — dependency and propagation

Status: proved

This certificate belongs to the critical layer. The P13R diplomatic transcription and its 78-page PDF are immutable. No wording below authorizes a silent change to that layer. Coordinates are physical PDF/JP2 coordinates from the frozen map.

Theorem. `DEPENDENCY_PROPAGATION_EDGES.csv` records every material downstream dependency of the 36 repairs and the four deferred proof audits.

Proof. Local typographical errors terminate inside their work. W05-SE002 points to W08-PE001; W05-SE001 points to the PDF051 classification. W09-SE001 feeds Proposition I and rational resolvents; W09-SE002 and W09-SE003 feed the normal-subgroup decomposition; W09-SE004 feeds the radical resolvent; W09-SE005–006 feed Propositions VII–VIII. The W10 index repairs feed the affine coordinate maps, while PE006 feeds the support trichotomy and DA003 independently blocks the terminal $p=3$ inference. Each edge is labelled by type and by one of the outcomes “survives,” “survives under hypotheses,” or “blocked.”

No later page silently inherits an unrepaired false statement. The diplomatic text remains historical evidence; the graph belongs to the critical layer.

1.7 POST-P13-A003 — historical and modern prior-notice search

Status: proved

This certificate belongs to the critical layer. The P13R diplomatic transcription and its 78-page PDF are immutable. No wording below authorizes a silent change to that layer. Coordinates are physical PDF/JP2 coordinates from the frozen map.

Method. The bounded corpus comprised the local 1846, 1897, and 1908 witnesses; Numdam and EuDML records; the independent ETH 1897 copy metadata; the Harvard-copy Wikisource transcription and revision record; the 1962 and 1976 Bourgne–Azra critical-edition records; Neumann’s 2011/2013 edition record; Dicker’s 2026 Proposition-II paper; and exact-phrase web searches logged in SEARCH_QUERY_LEDGER.csv.

Result. The 51-row BIBLIOGRAPHIC_PRIOR_NOTICE_MATRIX.csv distinguishes (i) explicit notice, (ii) a correct earlier witness without explicit erratum, (iii) a dated silent correction, (iv) an inaccessible candidate errata source, and (v) no prior notice found in the searched corpus. The 1976 printing is known from René Taton’s 1977 review to contain an errata list and two concordances, but its item-level errata were not accessible; it is therefore never used as proof that a particular error was previously noticed.

Conclusion. No record is called new. Negative results are bounded search results only.

1.8 POST-P13-A004 — PDF047 finite-field degree argument

Status: proved

Theorem. Under the dependencies printed in the argument, the allegedly larger degree μ cannot exceed ν ; in fact $\mu=\nu$.

Proof. Let the initial root i generate the finite field $F_{\{p^\mu\}}$. The construction writes the coefficients and hence the selected quantities in the field generated by i , so the field $F_{\{p^\nu\}}$ generated by the constructed element is a subfield of $F_{\{p^\mu\}}$. The finite-field subfield theorem gives $\nu|\mu$. Conversely, the text states that i is a rational function of the constructed quantity; therefore $F_{\{p^\mu\}}$ is a subfield of $F_{\{p^\nu\}}$, giving $\mu|\nu$. Hence $\mu=\nu$. The compressed source does not spell out both inclusions or invoke the subfield theorem, but its conclusion is correct.

Disposition. No new source-error ID is created. The critical layer supplies the omitted inclusions and divisibilities.

1.9 POST-P13-A005 — PDF051 converse and the degree-9/25 exceptions

Status: rejected

Theorem. The printed claim that the only exceptional solvable primitive degrees outside the one-dimensional semilinear form are 9 and 25 is false.

Counterexample. Let $W=F_2^2$ and $V=W\oplus W$. Let $A=[[0,1],[1,1]]$ of order 3. In $GL(V)$, set $g_1=\text{diag}(A,I)$, $g_2=\text{diag}(I,A)$, and let s interchange the two summands. Then $H=\langle g_1, g_2, s \rangle$ is $(C_3 \times C_3) \rtimes C_2$ of order 18 and is solvable. The two C_3 factors afford nonisomorphic irreducible two-dimensional modules on the two summands; an H -invariant subspace is invariant under their normal product and is therefore a sum of selected summands, while s interchanges the summands. Thus only 0 and V are invariant. Hence $G=V \rtimes H$ is a solvable primitive affine group of degree 16.

If this action were one-dimensional semilinear, H would embed in $\Gamma L(1,16)$, whose order is $(16-1) \cdot 4=60$. Lagrange’s theorem forbids an order-18 subgroup because 18 does not divide 60. Therefore degree 16 is a counterexample, distinct from 9 and 25. Exact enumeration in qa/MATHEMATICAL_CHECKS.json verifies $|H|=18$ and irreducibility.

Replacement theorem. A finite solvable primitive permutation group is affine: it has a regular elementary-abelian minimal normal subgroup V and is $V \rtimes H$ with H acting faithfully and irreducibly. Embedding H in $\Gamma L(1,p^d)$ is an additional condition, not a consequence of solvability and primitivity.

1.10 POST-P13-A006 — W08-PE001 and the p=11 lowering

Status: repaired

Theorem. Replacing the repeated 5 by 4 yields the corrected matching

$\{\infty, 0\}, \{1, 2\}, \{3, 6\}, \{4, 8\}, \{5, 10\}, \{9, 7\}$

on $P^1(F_{11})$. Its setwise stabilizer in $PSL(2, 11)$ has order 60 and index 11.

Proof. The twelve entries are now all distinct, so the six pairs partition the projective line. Exact enumeration of determinant-one fractional-linear transformations modulo scalar sign gives $|PSL(2, 11)|=660$. Testing the induced action on the six unordered pairs gives 60 stabilizing elements. The index is $660/60=11$, supplying the degree-11 resolvent/lowering structure. The 1846 publication prints 4; the 1897 edition alone duplicates 5.

Propagation. The correction restores the example promised by Liouville’s $p=11$ exception. It does not alter the diplomatic letter.

1.11 POST-P13-A007 — W08-PE002, équation/réduction

Status: proved

Collation. The 1846 publication reads “cette réduction n’est pas possible.” The 1897 edition reads “cette équation n’est pas possible.” Tannery’s 1908 collation explicitly reports the manuscript/Liouville reading and the later substitution.

Semantic certificate. The antecedent is the lowering of a modular equation from degree $p+1$ to degree p . “Réduction” names that operation. “Équation” instead makes the sentence say that the equation itself is impossible, which is not the mathematical assertion being discussed. The critical reading is therefore “réduction.”

Downstream effect. No displayed formula changes. The correction narrows the impossibility claim to the degree-lowering operation and aligns it with the $p=7$ and $p=11$ exceptions.

1.12 POST-P13-A008 — W08-WV001, Legendre relation

Critical-layer certificate. The frozen 78-page P13R diplomatic edition is immutable.

Status: repaired

Source replay. The frozen 1897 target and the 1846 publication print

$$FE' + EF' - FF' = \frac{\pi}{2}.$$

Tannery’s 1908 collation states that Galois wrote the same theorem in another form. Direct inspection of the 600 ppi TIFF master corrects a defect in the former GPT apparatus: the first letter of the manuscript formula has three horizontal strokes and is **E**, not **F**. Tannery prints

$$E'F'' - E''F' = \frac{\pi}{2}\sqrt{-1}.$$

The earlier apparatus form $F'F'' - E''F'$ is therefore rejected as a secondary-witness transcription error. This correction does not alter the 1897 diplomatic text.

Theorem (equivalence of the two Legendre forms). Put

$$F = K(k), \quad F_c = K(k'), \quad E = E(k), \quad E_c = E(k').$$

The printed relation is

$$EF_c + E_cF - FF_c = \frac{\pi}{2}.$$

Tannery 1908 formula replay

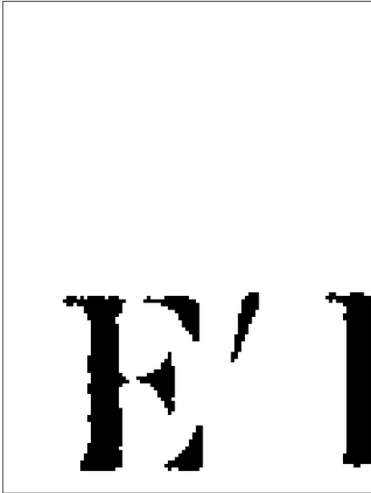
ne 31),

$$FE' + EF' - FF' = \frac{\pi}{2},$$

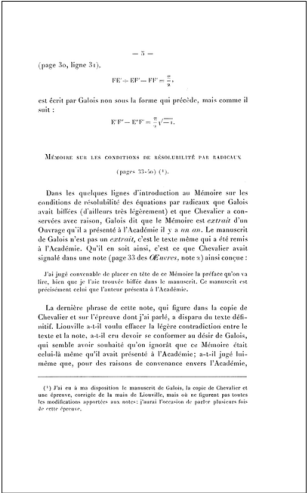
Galois non sous la forme qui précède, mais comme il

$$E'F'' - E''F' = \frac{\pi}{2}\sqrt{-1}.$$

First glyph of the manuscript formula



Full 600 ppi page context



The lower first letter is E: upper, middle, and lower horizontal strokes are present. Critical reading: $E'F'' - E''F' = (\pi/2)\sqrt{-1}$. The former $F'F''$ apparatus reading is rejected.

Figure 1.1: Tannery 1908 formula and character replay.

Choose the real/imaginary half-period pair and a compatible second-kind pair by

$$\omega_1 = F, \quad \omega_2 = iF_c, \quad \eta_1 = E, \quad \eta_2 = i(F_c - E_c).$$

Then

$$\begin{aligned} \eta_1\omega_2 - \eta_2\omega_1 &= E(iF_c) - i(F_c - E_c)F \\ &= i(EF_c + E_cF - FF_c) \\ &= \frac{\pi i}{2}. \end{aligned}$$

Identifying Tannery's manuscript symbols by

$$E' = \eta_1, \quad F'' = \omega_2, \quad E'' = \eta_2, \quad F' = \omega_1$$

gives exactly

$$E'F'' - E''F' = \frac{\pi i}{2}.$$

The chosen second-kind period vector is admissible, not merely an algebraic relabelling. Start with any second-kind differential having the same normalized principal part and period vector $(\tilde{\eta}_1, \tilde{\eta}_2)$. The Riemann bilinear relation gives the same determinant $\pi i/2$. Hence the difference between the desired vector and the initial vector has zero wedge product with $\omega = (\omega_1, \omega_2)$, so it is $c\omega$ for some constant c . Adding c times the holomorphic first-kind differential realizes exactly the desired period vector.

This identification is compatible with the usual freedom to add a multiple of a first-kind differential to a second-kind differential: replacing η_j by $\eta_j + c\omega_j$ leaves the determinant invariant, since

$$(\eta_1 + c\omega_1)\omega_2 - (\eta_2 + c\omega_2)\omega_1 = \eta_1\omega_2 - \eta_2\omega_1.$$

Thus the manuscript determinant and the 1846/1897 complete-integral formula express the same Legendre relation under an explicit half-period/quasi-period normalization. Reversing the orientation of the second cycle changes the signs of both second-cycle quantities; Tannery's displayed right-hand side fixes the orientation represented in his transcription.

Historical adjudication. Tannery 1908 is the earliest securely dated explicit notice located: he prints both forms and states that Galois wrote the theorem in the determinant form. DLMF 19.7.1 supplies the modern complete-integral identity. Beppo Levi's 1927 paper is explicitly devoted to the half-period/quasi-period determinant $\eta_1\omega_2 - \eta_2\omega_1 = \pi i/2$, confirming that the determinant normalization is a standard Legendre relation rather than a different theorem.

Propagation. Neither formula is used in a later deduction in the 1897 letter. The repair closes the witness-variant task and corrects the critical apparatus only; it has no downstream change to the diplomatic edition or to later Galois proofs.

1.13 POST-P13-A009 — W09-PE001 and W09-PE002

Status: proved

Collation. At PDF065 the 1897 edition prints “celle fonction,” whereas the 1846 publication prints “cette fonction.” At PDF071 the 1897 edition prints “propriété.” The local 1846 collation did not establish a comparison reading for that exact token, and no manuscript-collation source supplies a competing substantive reading.

Conclusion. Both are textual errors: the former is a demonstrative-agreement error supported by a correct 1846 witness; the latter is a missing diacritic established from the 1897 spelling and French orthography. The critical layer restores “cette fonction” and “propriété,” but records distinct provenance classes. Exact-phrase and edition searches found no explicit itemized notice; the matrix makes no novelty claim.

1.14 POST-P13-A010 — W09-SE001 and Lemma III

Status: repaired

Theorem (Bézout repair). Let K be the coefficient field and V the separating resolvent. Suppose $P(X)$ and $Q(X,V)$ in $K(V)[X]$ have exactly one common root a_i in a splitting field and all relevant roots are separable. Then a_i belongs to $K(V)$.

Proof. The monic greatest common divisor $D(X)=\gcd(P,Q)$ has precisely the common roots, hence $D(X)=X-a_i$. The Euclidean algorithm in the PID $K(V)[X]$ gives $U(X)P(X)+W(X)Q(X,V)=X-a_i$. Comparing constant terms, or evaluating the identity in the quotient algebra, expresses a_i as an element of $K(V)$. Thus each root singled out by the resolvent is rational in V .

Propagation. Proposition I's recovery of the roots from V is now proved. Every later substitution or resolvent construction that treats the roots as rational functions of V inherits this theorem. Tannery's 1908 collation reports Galois's own judgment that the abbreviated proof was insufficient.

1.15 POST-P13-A011 — W09-SE002 and W09-SE003

Status: repaired

Theorem (intersection/index). Let L/K be the original splitting field, $M = K(r)$ an auxiliary extension, $E = L \cap M$, $G = \text{Gal}(L/K)$, and $H = \text{Gal}(L/E)$. Then $[G : H] = [E : K]$ and $[E : K] \mid [M : K]$.

Proof. The Galois correspondence gives $[G : H] = [E : K]$. Since E is a subfield of M , the tower law gives $[M : K] = [M : E][E : K]$. If $[M : K] = p$ is prime, the index is 1 or p . Without prime degree the dichotomy is false: $r = \sqrt{2} + \sqrt{3}$ has degree 4, contains $\mathbb{Q}(\sqrt{2})$, and reduces its C_2 splitting group with index 2.

Theorem (Proposition III normality). If M/K is also a splitting field, then E/K is finite Galois, so H is normal in G . Consequently the subgroups arising from conjugate auxiliary roots coincide after all auxiliary roots are adjoined.

Propagation. These two theorems replace the undefined p and the omitted proof. They control all dependent group decompositions and the subsequent radical tower.

1.16 POST-P13-A012 — W09-SE004 Fourier coefficient

Status: repaired

Theorem. Let $\theta_0, \dots, \theta_{p-1}$ be a nonconstant cyclic orbit and ζ a primitive p -th root of unity. At least one nontrivial Fourier coefficient $R_m = \sum_j \zeta^{mj} \theta_j$, $1 \leq m \leq p-1$, is nonzero.

Proof. Fourier inversion gives $\theta_j = p^{-1} \sum_m \zeta^{-mj} R_m$. If every R_m for $m \neq 0$ vanished, every θ_j would equal $p^{-1} R_0$, contradicting nonconstancy. The printed choice R_1 need not work: for $p=3$ and $\theta_j = 1 + \omega^j t$, one has $R_1 = 0$ but $R_2 = 3t \neq 0$.

A cyclic shift multiplies R_m by a p -th root of unity, so R_m^p is invariant. Choosing a nonzero nontrivial coefficient therefore supplies the intended radical adjunction. The radical-tower conclusion survives with this selection rule.

1.17 POST-P13-A013 — W09-SE005, W09-SE006, Propositions VII-VIII

Status: repaired

Counterexample to the unqualified reducibility claim. The irreducible polynomial X^3-2 becomes reducible over $\mathbb{Q}(\sqrt[3]{2})$ but does not split; the residual quadratic has negative discriminant in the real embedding. Thus reducibility does not force the group to become trivial in an arbitrary radical extension.

Kummer repair. Let K contain the relevant p -th roots of unity and let M/K be an irreducible cyclic Galois Kummer extension of prime degree p . For an irreducible polynomial of prime degree n with splitting field L , put $H = \text{Gal}(L/L \cap M)$. Then H is normal. Its orbits form a block system in a prime-degree transitive action, so H is transitive or trivial. If the polynomial becomes reducible over M , H is not transitive and is therefore trivial; the polynomial splits completely. This is the context in which Propositions VI-VII survive.

Affine repair. In Proposition VIII insert “nonidentity.” The identity $k \mapsto k$ fixes every letter; a nonidentity affine map $k \mapsto ak+b$ fixes at most one. Hence adjoining two roots kills every nonidentity group element, and the final affine necessity/sufficiency argument survives.

1.18 POST-P13-A014 — W09 provenance corpus

Status: proved

This certificate belongs to the critical layer. The P13R diplomatic transcription and its 78-page PDF are immutable. No wording below authorizes a silent change to that layer. Coordinates are physical PDF/JP2 coordinates from the frozen map.

The six W09 witness variants are all explicitly grounded in Tannery’s 1908 collation: omitted definitions, Lemma III’s insufficiency, deletion of the prime-degree phrase, the revised Proposition III, a marginal construction sentence, and the expanded Cauchy citation. The 1846 witness supplies the correct form for W09-PE001; W09-PE002 is diagnosed from the 1897 spelling itself, without a securely established 1846 comparison token. The 1962 critical edition is bibliographically verified; a 1977 review verifies that the 1976 facsimile printing added supplementary pages containing an errata list and two concordances, but the item-level entries were inaccessible. Neumann’s 2011 book has a corrected second printing dated September 2013. Dicker’s paper submitted July 22, 2026 specifically reconstructs Proposition II, not Lemma III or Proposition III.

The matrix records these as established prior art where direct evidence exists. W09-SE004–006 are recorded as “no prior notice found in the searched corpus,” not as new results.

1.19 W10-DA001 — prime-power theorem and unique blocks

Status: repaired

Theorem. A finite solvable primitive permutation group G has degree p^d and embeds in $\text{AGL}(d,p)$.

Proof. Choose a minimal nontrivial normal subgroup N of G . Solvability makes N elementary abelian of order p^d . The N -orbits form a G -invariant block system; primitivity makes N transitive. Since N is abelian, a point stabilizer N_α is normal in G : all point stabilizers in N are equal by abelianness and their intersection is the kernel of the faithful action, hence trivial. Thus N is regular. Identifying the points with N gives an affine space over F_p . A point stabilizer H acts faithfully by conjugation on N and is irreducible, because an H -invariant proper nonzero subgroup of N would yield a nontrivial block system. Hence $G = N \rtimes H \leq \text{AGL}(d,p)$.

Disposition. The printed unique-block construction is incomplete but its prime-power conclusion survives under the inherited finite solvable primitive hypotheses.

1.20 W10-DA002 — affine linearity

Status: repaired

Qualified theorem. Under the inherited hypotheses that the permutation group is finite, solvable, and primitive of degree p^d , W10-DA001 identifies the letters with the regular elementary-abelian socle V and identifies the point stabilizer with an irreducible subgroup of $\text{GL}(V)$. Every group element is therefore affine: $x \mapsto Ax+b$.

Failure of the unqualified reading. Primitivity alone does not imply affine linearity. For example, $S_{\{p^2\}}$ in its natural action is primitive, but for $p^2 \geq 5$ it has nonabelian simple socle $A_{\{p^2\}}$ and no regular elementary-abelian normal subgroup. It cannot be an affine group of degree p^2 .

Conclusion. Formula (A), after W10-PE005 is repaired, is valid only inside the inherited solvable-primitive argument.

1.21 W10-DA003 — PDF089 conjuguées and pairwise equality

Status: rejected

Historical finding. Cauchy’s technical phrase “système de substitutions conjuguées” designated what modern historians translate as a “system of conjoined substitutions,” namely a substitution system closed under composition; Hollings explicitly warns that the translation “conjugate substitutions” is best avoided because of its modern group-theoretic meaning. In the PDF089 sentence, the immediately preceding clause already says that the substitutions are transformed into one another by the translations $((k, k+m))$. Thus the historically supportable force is membership in, or transport within, the relevant closed substitution system/orbit—not pairwise commutativity. The lexical evidence does not establish a stronger relation.

Exact counterexample in the printed projective family. On $P^1(F_7)$, set $T_m(k) = m + 1/(k-m)$, with the usual values at infinity. Translation τ_1 conjugates T_0 to T_1 . Nevertheless direct evaluation shows $T_0 T_1 \neq T_1 T_0$. Thus even two members related by the evident translation conjugacy fail to commute.

Consequence. Neither the historically attested “conjoined-system” sense nor modern element conjugacy implies pairwise commutativity. The premise used at PDF089 is therefore unsupported; under modern conjugacy it is explicitly false. The terminal $p=3$ conclusion depending on it is blocked. No diplomatic wording is changed.

1.22 W10-DA004 — propagation through the p -power and p^2 arguments

Status: repaired

W10-PE001–PE004 restore the coordinate tuples and maps on PDF082–083. W10-PE005 restores the two-dimensional affine formula; with W10-DA001–002 it expresses the solvable primitive group as an affine group. W10-SE001 adds the nonidentity qualification to the fixed-space dimension count; its divisibility conclusion survives. W10-SE002 changes only the count of fixed-point-free nonidentity scalars from $p^2 - 1$ to $p^2 - 2$; the scalar subgroup order $p^2 - 1$ and total affine group order $p^2(p^2 - 1)(p^2 - p)$ survive. W10-PE006 restores the projective fixed-point equation and makes the printed discriminant correct, so the support-size trichotomy survives.

The final PDF089 pairwise-commutativity premise does not survive, by W10-DA003. Accordingly, the $p=3$ terminal inference is not carried into the corrected theorem set.

1.23 W11-DA001 — table-of-contents 1 to v

Status: proved

The frozen page map and the primary 1897 images place Picard’s Introduction on roman folios v-x. The 1897 table of contents nevertheless prints Arabic 1. An independent Harvard-copy Wikisource transcription prints v; its permanent page record states that the page was last modified on July 22, 2018. This is the earliest securely dated correction located in the accessible corpus.

The official ETH-Bibliothek copy, shelfmark Rar 4575 and DOI 10.3931/e-rara-19915, was subsequently retrieved at page level. Its table of contents independently prints the same Arabic 1, showing that the error belongs to the 1897 printed edition rather than to the canonical scan alone. The 1976 Bourgne–Azra printing is known to add an errata list, but that item-level list was not accessible, so no earlier notice is attributed to it.

1.24 W11-DA002 — printer job number

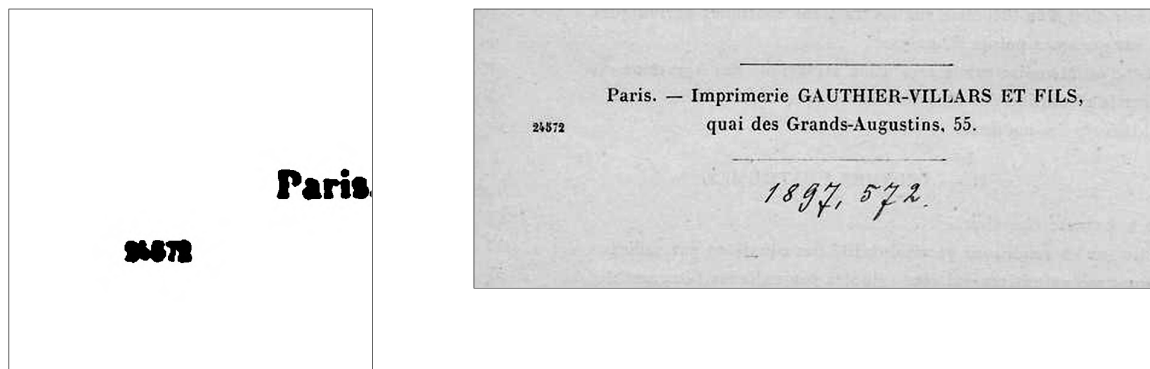
Status: proved

Independent witness. The official ETH-Bibliothek digitization of the independent 1897 copy, shelfmark Rar 4575 and DOI 10.3931/e-rara-19915, was retrieved from the library’s public download endpoint. Its PDF page 79 contains the colophon. The small printer number to the left of the second line is visibly

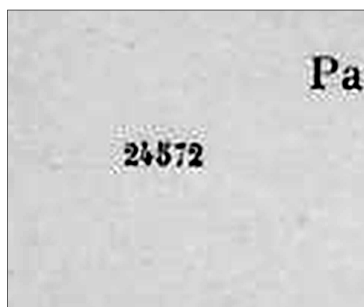
[24572.]

It has five digits, not four. The official ETH OCR independently transcribes 24572 on the same colophon page.

Canonical 1897 copy: fused job-number region Independent ETH Rar 4575 copy, PDF page 79



Native ETH digit crop



Resolved reading: 24572, consisting of five digits 2-4-5-7-2. The official ETH OCR independently gives 24572. The earlier four-digit premise is rejected.

Figure 1.2: Canonical and ETH colophon comparison.

Character-by-character certificate. In the independent image, the first and fifth glyphs have the curved upper stroke, diagonal descent, and basal stroke of 2; the second has the vertical and cross-stroke structure of 4; the third has the top bar and lower bowl of 5; and the fourth has a top bar with descending diagonal, hence 7. The sequence is therefore 2-4-5-7-2. The canonical JP2 remains locally fused, but its five dark glyph components are consistent with the independent reading. The independent image, rather than the OCR alone, is decisive.

Disposition. W11-U001 is closed in the critical layer with reading 24572. The historical 1897 diplomatic page is not re-typeset or silently altered; the critical appendix and machine-readable resolution ledger carry the reading and evidence. No Gauthier-Villars printer register was needed after recovery of the independent comparison copy.

1.25 W11-DA003 — terminal topology and P10/P11 boundary

Status: proved

Replay of the primary PDF, JP2 leaves, disposition map, and integrated candidate gives the following exact topology: PDF091/L0090 is the predecessor blank owned by W10; PDF092/L0091 is the historical table of contents; PDF093/L0092 is the colophon; PDF094/L0093, PDF095/L0094, and PDF096/L0095 are three distinct trailing blanks. The P10/P11 boundary introduces no text join. The corrected critical page reference v does not alter the historical table in the diplomatic layer.

All six physical leaves remain dispositioned exactly once. The one-click reader preserves the frozen 78-page diplomatic PDF as its prefix; the critical appendix follows it and does not collapse any blank topology.

1.26 Resolved after the R2 deep replay

Two records formerly marked `unresolved_after_bounded_search` are now closed:

record_id	R2 status	resolution
W08-WV001 / POST-P13-A008	repaired	The first manuscript glyph is E, and the determinant form is proved equivalent to the printed Legendre relation under an explicit period normalization.
W11-U001 / W11-DA002	proved	The independent ETH 1897 copy and OCR give the five-digit printer job number 24572.

1.27 Open readings after the R2 deep replay

Three records remain open. They are inherited W01 image/provenance questions outside the mathematical 21-task scope; neither post-P13 task remains unresolved.

record_id	status	missing evidence
W01-U001	carried forward unresolved outside 21-task scope	A character-resolved independent image or documented portrait-plate source sufficient to read the faint lower-left signature on PDF019.
W01-U002	carried forward unresolved outside 21-task scope	Independent provenance evidence sufficient to adjudicate the relation between the degraded PDF021 portrait and PDF019 without silently deduplicating witnesses.
W01-U003	carried forward unresolved outside 21-task scope	A higher-fidelity device image or printer-source record sufficient to expand the publisher-device ribbon legend character by character.

1.28 Historical-priority conclusions

Direct evidence establishes Liouville's 1846 footnote as notice of the $p=7$ and $p=11$ modular exceptions; Tannery's 1908 collation as explicit notice of multiple W08-W10 editorial and proof defects; the 1976 Bourgne-Azra printing as containing an added errata list whose item-level contents were unavailable; and the Harvard-copy Wikisource page as a dated silent correction of the contents reference to roman v on July 22, 2018. All other negative findings are stated only as "no prior notice found in the searched corpus."

1.29 Bibliographic sources

- **SRC-LOCAL-1897.** Évariste Galois; Société mathématique de France; introduction by Émile Picard, *Œuvres mathématiques d'Évariste Galois* (1897), Paris: Gauthier-Villars et Fils, physical PDF 1-96; printed x, 1-61 plus back matter.
- **SRC-LOCAL-1846.** Évariste Galois; edited by Joseph Liouville, *Œuvres mathématiques* (1846), Journal de Mathématiques Pures et Appliquées, série 1, vol. 11, 381-444; local PDF 1-65. https://www.numdam.org/item/JMPA_1846_1_11__381_0/
- **SRC-LOCAL-1908.** Jules Tannery, editor, *Manuscrits d'Évariste Galois* (1908), Paris: Gauthier-Villars, local PDF 1-77; Commons scan 87 leaves. https://commons.wikimedia.org/wiki/File:Galois_-_Manuscrits,_édition_Tannery,_1908.djvu
- **SRC-NUMDAM-1846.** Numdam, *Numdam record for the 1846 publication* (1846/online), Numdam item: JMPA 1846, series 1, volume 11, page 381, 381-444; record marks corrected-by Errata. https://www.numdam.org/item/JMPA_1846_1_11__381_0/

- **SRC-ETH-1897.** ETH-Bibliothek Zürich, *ETH-Bibliothek independent copy of Oeuvres mathématiques* (1897; online 2013-10-08), Rar 4575; DOI 10.3931/e-rara-19915, X, 61 pages, one leaf. <https://www.e-rara.ch/zut/content/titleinfo/6262819>
- **SRC-WIKI-1897.** French Wikisource contributors, *Harvard College Library 1897 copy transcription* (1897 scan; current online transcription), Gauthier-Villars 1897, complete index. https://fr.wikisource.org/wiki/Livre:Galois_-_Œuvres_mathématiques,_Gauthier-Villars,_1897.djvu
- **SRC-WIKI-TOC.** French Wikisource contributors, *1897 table-of-contents transcription, page 91* (last modified 2018-07-22), oldid 7722271, page 91. https://fr.wikisource.org/w/index.php?title=Page:Galois_-_Œuvres_mathématiques,_Gauthier-Villars,_1897.djvu/91&oldid=7722271
- **SRC-DLMF-LEGENDRE.** NIST Digital Library of Mathematical Functions, *DLMF §19.7(i), Legendre's Relation* (current reference), Equation 19.7.1, §19.7(i). <https://dlmf.nist.gov/19.7.E1>
- **SRC-BDIM-LEGENDRE-1927.** Beppo Levi, *Sulla relazione $\eta_1\omega_2 - \eta_2\omega_1 = \pi i/2$ nella teoria delle funzioni ellittiche* (1927), *Bollettino dell'Unione Matematica Italiana*, series 1, vol. 6, no. 3, pp. 137-141. https://www.bdim.eu/item?id=BUMI_1927_1_6_3_137_0
- **SRC-BOURGNE-AZRA-1962.** Robert Bourgne and Jean-Pierre Azra, editors, *Écrits et mémoires mathématiques d'Évariste Galois* (1962), Gauthier-Villars; xxii+541 pp., critical edition. <https://doi.org/10.1017/S0008439500026989>
- **SRC-BOURGNE-AZRA-1976.** Robert Bourgne and Jean-Pierre Azra, editors, *Écrits et mémoires mathématiques d'Évariste Galois, 2e éd.* (1976), new facsimile printing with supplementary errata and two concordances, xxxi+541 pp.. https://www.persee.fr/doc/rhs_0151-4105_1977_num_30_2_1486
- **SRC-NEUMANN-2013.** Peter M. Neumann, *The mathematical writings of Évariste Galois* (2011; corrected second printing September 2013), EMS Heritage of European Mathematics 104, 1-410. <https://ems.press/books/hem/102/contents>
- **SRC-DICKER-2026.** Math Dicker, *The Significance of Proposition II in Galois' Mémoire, The Origin of Galois Automorphisms* (2026-07-22), arXiv:2607.20147, abstract and paper. <https://arxiv.org/abs/2607.20147>
- **SRC-CAUCHY-TERMS.** Christopher D. Hollings, *'Nobody could possibly misunderstand what a group is': a study in early twentieth-century group axiomatics* (2017), *Archive for History of Exact Sciences* 71, 409–481, especially §2.1 on early substitution terminology. <https://doi.org/10.1007/s00407-017-0193-8>
- **SRC-AFFINE-PRIMITIVE.** Attila Maróti and Saveliy V. Skresanov, *Bounds for the Diameters of Orbital Graphs of Affine Groups* (2023), *Vietnam Journal of Mathematics* 51, 617–631; introductory definition of affine primitive groups. <https://doi.org/10.1007/s10013-023-00607-5>
- **SRC-LI-2003.** Cai Heng Li, *The Finite Primitive Permutation Groups Containing an Abelian Regular Subgroup* (2003), *Proc. London Math. Soc.* 87, 725-747. <https://doi.org/10.1112/S0024611503014266>

1.30 Machine-readable companions

The authoritative structured companions are MASTER_21_TASK_LEDGER.csv, CRITICAL_ERRATA_CATALOGUE.csv, BIBLIOGRAPHIC_PRIOR_NOTICE_MATRIX.csv, SEARCH_QUERY_LEDGER.csv, DEPENDENCY_PROPAGATION_EDGES.csv, the frozen and R2 resolution/open ledgers, and qa/MATHEMATICAL_CHECKS.json.